

Multi-view traffic videos

Overhead view



Vehicle view (front)



Phase timeline



Actor tracks



Domain setting



Phase-Aware Evidence Construction

1 Phase clip extraction



phase clips



2 View selection



role/view metadata



3 Track summarization



normalized box summaries
 $[x_1, y_1, x_2, y_2]$
on 0-1000 grid

Track-Grounded Prompt Builder

Structured prompt template
(checklist slots)

- ☐ location
- ☐ attention
- ☐ behavior
- ☐ context attributes

Input fields

- phase label
- actor role
- camera view
- bbox track summary

Transfers structured state
(evidence, roles, tracks),
not synthetic appearance.

Synthetic VLM Adaptation

Video-Language Model
+ LoRA



Caption
SFT



VQA
SFT



trained only on
synthetic data

Captioning

Caption
Candidate
Generation



Fact-Locked
Caption
Fusion



Role/Phase
Gating



Pedestrian
caption

Vehicle
caption

consistency
(grounding)

VQA

Evidence-Diverse
VQA Agents



Question-Type
Routing



Consensus
Selection



Question groups

attribute

geometry

visibility

action

environment

Official Track 2 Submission

caption_
pedestrian

caption_
vehicle

VQA
answers